CSE421 : Computer Networks
Lab Project
Group No : 2605

Group Members: [24141166,ADNAN SAMI SRIJON], [24241169,RUBAIYA RAHIM], [24241170,MD. SAMSUL ARAFIN ANAN], [24241171,MAIMUNA ANJUM]

Smart Integrated Service Network (SISN)

This project presents the design and implementation of a Smart Integrated Service Network (SISN), a complete enterprise network built using Cisco Packet Tracer. The network interconnects six organizational units, enabling efficient and reliable communication through a combination of static and dynamic routing techniques, automated IP address assignment, and server-based services.

The base network address for this implementation is 9.9.0.0/16, derived from the fourth group member's student ID (24241171), where the sum of the last three digits (1+7+1=9) forms the first two octets of the network address.

VLSM Tree

4th member id = 24241171

sum = 1 + 7 + 1 = 9

Base network = 9.9.0.0/16

Given,

HO - /23

TC - /24

SL - /25

SD - /25

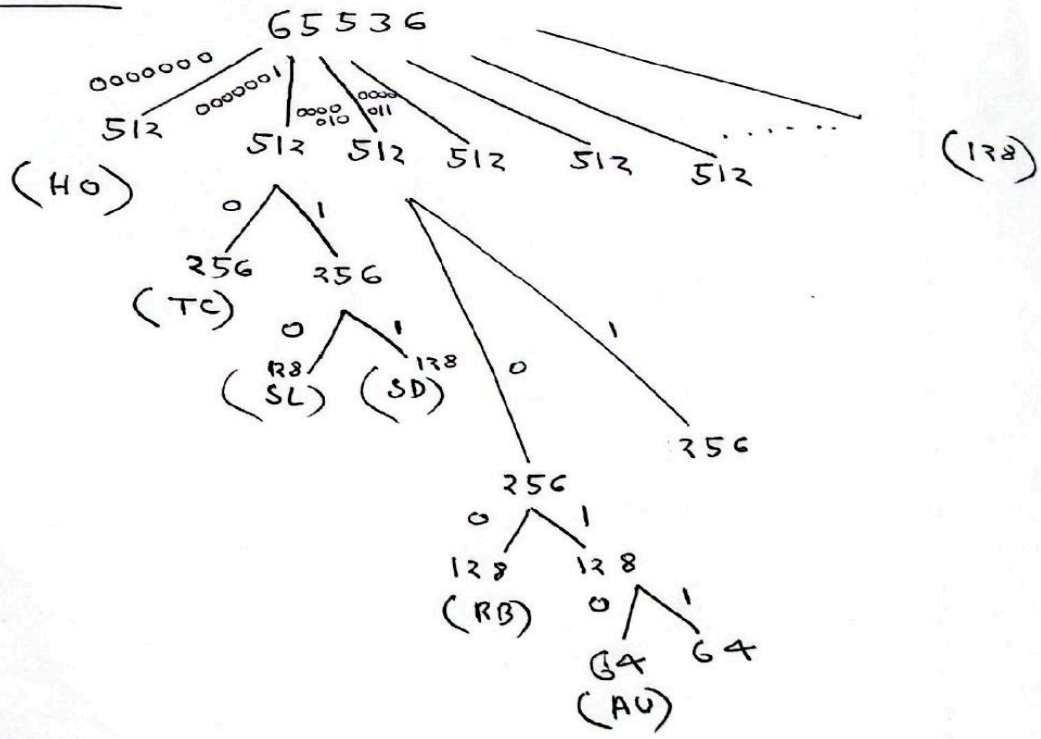
RB - /25

AU - /26

$2^{16} = 65536$

$2^9 = 512$

VLSM Tree -



Network address,

HO - 9.9.00000000.00000000 /23

9.9.0.0

TC - 9.9.00000010.00000000 /24

9.9.2.0/24

SL - 9.9.00000011.00000000 /25

9.9.3.0/25

SD - 9.9.00000011.10000000 /25

9.9.3.128 /25

RB - 9.9.00000100.00000000 /25

9.9.4.0/25

AU - 9.9.00000100.10000000 /26

9.9.4.128 /26

CamScanner

Router configuration commands (CLI Commands)

HO-Router

```
enable
configure terminal
hostname HO-Router

interface GigabitEthernet0/0
ip address 9.9.0.1 255.255.254.0
no shutdown
exit

interface Serial0/3/0
ip address 9.9.4.193 255.255.255.252
clock rate 64000
no shutdown
exit

interface Serial0/3/1
ip address 9.9.4.197 255.255.255.252
clock rate 64000
no shutdown
exit

interface Serial0/2/0
ip address 9.9.4.201 255.255.255.252
clock rate 64000
no shutdown
exit

ip route 9.9.2.0 255.255.255.0 9.9.4.194

router rip
version 2
no auto-summary
network 9.9.0.0
network 9.9.4.0
redistribute static
exit

end
write memory
```

TC-Router

```
enable
configure terminal
hostname TC-Router

interface GigabitEthernet0/0
ip address 9.9.2.1 255.255.255.0
no shutdown
exit
```

```
interface Serial0/3/0
ip address 9.9.4.194 255.255.255.252
no shutdown
exit

ip route 0.0.0.0 0.0.0.0 9.9.4.193

ip dhcp excluded-address 9.9.2.1 9.9.2.10
ip dhcp pool TC-POOL
network 9.9.2.0 255.255.255.0
default-router 9.9.2.1
dns-server 9.9.0.6
exit

end
write memory
```

SL-Router

```
enable
configure terminal
hostname SL-Router

interface GigabitEthernet0/0
ip address 9.9.3.1 255.255.255.128
no shutdown
exit

interface GigabitEthernet0/1
no shutdown
exit

interface Serial0/3/0
ip address 9.9.4.198 255.255.255.252
no shutdown
exit

interface Serial0/3/1
ip address 9.9.4.205 255.255.255.252
clock rate 64000
no shutdown
exit

ip route 9.9.2.0 255.255.255.0 9.9.4.197

router rip
version 2
no auto-summary
network 9.9.3.0
network 9.9.4.0
exit
```

```
end
write memory
```

SD-Router

```
enable
configure terminal
hostname SD-Router

interface GigabitEthernet0/0
ip address 9.9.3.129 255.255.255.128
no shutdown
exit

interface GigabitEthernet0/1
no shutdown
exit

interface Serial0/3/0
ip address 9.9.4.206 255.255.255.252
no shutdown
exit

interface Serial0/3/1
ip address 9.9.4.213 255.255.255.252
clock rate 64000
no shutdown
exit

ip route 9.9.2.0 255.255.255.0 9.9.4.205

router rip
version 2
no auto-summary
network 9.9.3.128
network 9.9.4.0
exit

end
write memory
```

RB-Router

```
enable
configure terminal
hostname RB-Router

interface GigabitEthernet0/0
ip address 9.9.4.1 255.255.255.128
no shutdown
exit

interface GigabitEthernet0/1
```

```
no shutdown
exit

interface Serial0/3/0
ip address 9.9.4.202 255.255.255.252
no shutdown
exit

interface Serial0/3/1
ip address 9.9.4.217 255.255.255.252
clock rate 64000
no shutdown
exit

ip route 0.0.0.0 0.0.0.0 Serial0/3/0

router rip
version 2
no auto-summary
network 9.9.4.0
exit

ip dhcp excluded-address 9.9.4.1 9.9.4.10
ip dhcp pool RB-POOL
network 9.9.4.0 255.255.255.128
default-router 9.9.4.1
dns-server 9.9.0.6
exit

end
write memory
```

AU-Router

```
enable
configure terminal
hostname AU-Router

interface GigabitEthernet0/0
ip address 9.9.4.129 255.255.255.192
no shutdown
exit

interface Serial0/3/0
ip address 9.9.4.214 255.255.255.252
no shutdown
exit

interface Serial0/3/1
ip address 9.9.4.218 255.255.255.252
no shutdown
exit
```

```
router rip
version 2
no auto-summary
network 9.9.4.128
network 9.9.4.0
exit

ip route 0.0.0.0 0.0.0.0 Serial0/3/1 121

end
write memory
```

Network Documentation

1. VLSM Subnetting Table

Base Network: 9.9.0.0/16 | 4th Member ID: 24241171 | Sum of last 3 digits: 1+7+1 = 9

Unit	IP Address	Subnet Mask	Usable Range	Broadcast	Hosts
HO	9.9.0.0/23	255.255.254.0	9.9.0.1 – 9.9.1.254	9.9.1.255	510
TC	9.9.2.0/24	255.255.255.0	9.9.2.1 – 9.9.2.254	9.9.2.255	254
SL	9.9.3.0/25	255.255.255.128	9.9.3.1 – 9.9.3.126	9.9.3.127	126
SD	9.9.3.128/25	255.255.255.128	9.9.3.129 – 9.9.3.254	9.9.3.255	126
RB	9.9.4.0/25	255.255.255.128	9.9.4.1 – 9.9.4.126	9.9.4.127	126
AU	9.9.4.128/26	255.255.255.192	9.9.4.129 – 9.9.4.190	9.9.4.191	62

2. Inter-Router Link Subnets (/30)

Link	Network	Router A Interface	Router A IP	Router B Interface	Router B IP
HO ↔ TC	9.9.4.192/30	HO Serial0/3/0	9.9.4.193	TC Serial0/3/0	9.9.4.194
HO ↔ SL	9.9.4.196/30	HO Serial0/3/1	9.9.4.197	SL Serial0/3/0	9.9.4.198
HO ↔ RB	9.9.4.200/30	HO Serial0/2/0	9.9.4.201	RB Serial0/3/0	9.9.4.202
SL ↔ SD	9.9.4.204/30	SL Serial0/3/1	9.9.4.205	SD Serial0/3/0	9.9.4.206
SD ↔ AU	9.9.4.212/30	SD Serial0/3/1	9.9.4.213	AU Serial0/3/0	9.9.4.214
RB ↔ AU	9.9.4.216/30	RB Serial0/3/1	9.9.4.217	AU Serial0/3/1	9.9.4.218

3. IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Gateway	Type
HO-Router	GigabitEthernet0/0	9.9.0.1	255.255.254.0	—	Static
HO-Router	Serial0/3/0	9.9.4.193	255.255.255.252	—	Static
HO-Router	Serial0/3/1	9.9.4.197	255.255.255.252	—	Static
HO-Router	Serial0/2/0	9.9.4.201	255.255.255.252	—	Static
HO-PC1	FastEthernet0	9.9.0.2	255.255.254.0	9.9.0.1	Static
HO-PC2	FastEthernet0	9.9.0.3	255.255.254.0	9.9.0.1	Static
HO-Printer	FastEthernet0	9.9.0.4	255.255.254.0	9.9.0.1	Static
HO-EmailServer	FastEthernet0	9.9.0.5	255.255.254.0	9.9.0.1	Static
HO-DNS	FastEthernet0	9.9.0.6	255.255.254.0	9.9.0.1	Static
HO-WebServer	FastEthernet0	9.9.0.7	255.255.254.0	9.9.0.1	Static
TC-Router	GigabitEthernet0/0	9.9.2.1	255.255.255.0	—	Static
TC-Router	Serial0/3/0	9.9.4.194	255.255.255.252	—	Static
TC-PC1	FastEthernet0	DHCP (9.9.2.11+)	255.255.255.0	9.9.2.1	DHCP
TC-PC2	FastEthernet0	DHCP (9.9.2.12+)	255.255.255.0	9.9.2.1	DHCP
TC-Printer	FastEthernet0	DHCP (9.9.2.13+)	255.255.255.0	9.9.2.1	DHCP
TC-EmailServer	FastEthernet0	DHCP (9.9.2.14+)	255.255.255.0	9.9.2.1	DHCP
SL-Router	GigabitEthernet0/0	9.9.3.1	255.255.255.128	—	Static
SL-Router	Serial0/3/0	9.9.4.198	255.255.255.252	—	Static
SL-Router	Serial0/3/1	9.9.4.205	255.255.255.252	—	Static
SL-DHCPServer	FastEthernet0	9.9.3.2	255.255.255.128	9.9.3.1	Static
SL-PC1	FastEthernet0	DHCP (9.9.3.10+)	255.255.255.128	9.9.3.1	DHCP
SL-PC2	FastEthernet0	DHCP (9.9.3.11+)	255.255.255.128	9.9.3.1	DHCP
SL-Printer	FastEthernet0	DHCP (9.9.3.12+)	255.255.255.128	9.9.3.1	DHCP
SL-EmailServer	FastEthernet0	DHCP (9.9.3.13+)	255.255.255.128	9.9.3.1	DHCP
SD-Router	GigabitEthernet0/0	9.9.3.129	255.255.255.128	—	Static
SD-Router	Serial0/3/0	9.9.4.206	255.255.255.252	—	Static
SD-Router	Serial0/3/1	9.9.4.213	255.255.255.252	—	Static
SD-DHCPServer	FastEthernet0	9.9.3.130	255.255.255.128	9.9.3.129	Static
SD-PC1	FastEthernet0	DHCP (9.9.3.140+)	255.255.255.128	9.9.3.129	DHCP
SD-PC2	FastEthernet0	DHCP (9.9.3.141+)	255.255.255.128	9.9.3.129	DHCP
SD-Printer	FastEthernet0	DHCP (9.9.3.142+)	255.255.255.128	9.9.3.129	DHCP
SD-EmailServer	FastEthernet0	DHCP (9.9.3.143+)	255.255.255.128	9.9.3.129	DHCP
RB-Router	GigabitEthernet0/0	9.9.4.1	255.255.255.128	—	Static
RB-Router	Serial0/3/0	9.9.4.202	255.255.255.252	—	Static
RB-Router	Serial0/3/1	9.9.4.217	255.255.255.252	—	Static
RB-PC1	FastEthernet0	DHCP (9.9.4.11+)	255.255.255.128	9.9.4.1	DHCP
RB-PC2	FastEthernet0	DHCP (9.9.4.12+)	255.255.255.128	9.9.4.1	DHCP

RB-Printer	FastEthernet0	DHCP (9.9.4.13+)	255.255.255.128	9.9.4.1	DHCP
RB-EmailServer	FastEthernet0	DHCP (9.9.4.14+)	255.255.255.128	9.9.4.1	DHCP
AU-Router	GigabitEthernet0/0	9.9.4.129	255.255.255.192	—	Static
AU-Router	Serial0/3/0	9.9.4.214	255.255.255.252	—	Static
AU-Router	Serial0/3/1	9.9.4.218	255.255.255.252	—	Static
AU-PC1	FastEthernet0	9.9.4.130	255.255.255.192	9.9.4.129	Static
AU-PC2	FastEthernet0	9.9.4.131	255.255.255.192	9.9.4.129	Static
AU-Printer	FastEthernet0	9.9.4.132	255.255.255.192	9.9.4.129	Static
AU-EmailServer	FastEthernet0	9.9.4.133	255.255.255.192	9.9.4.129	Static

4. DHCP Configuration Summary

Unit	Method	Server / Router	Pool Name	Start IP	Excluded IPs
HO	Static	—	—	—	All devices static
TC	Router	TC-Router	TC-POOL	9.9.2.11	9.9.2.1 – 9.9.2.10
SL	Dedicated Server	SL-DHCPServer (9.9.3.2)	SL-POOL	9.9.3.10	9.9.3.1 – 9.9.3.9
SD	Dedicated Server	SD-DHCPServer (9.9.3.130)	SD-POOL	9.9.3.140	9.9.3.129 – 9.9.3.139
RB	Router	RB-Router	RB-POOL	9.9.4.11	9.9.4.1 – 9.9.4.10
AU	Static	—	—	—	All devices static

5. Server Configuration Summary

Server	Unit	IP Address	Service	Domain/URL	Protocol
HO-DNS	HO	9.9.0.6	DNS	www.sisn.local	DNS (Port 53)
HO-WebServer	HO	9.9.0.7	HTTP Web Server	www.sisn.local	HTTP (Port 80)
HO-EmailServer	HO	9.9.0.5	Email	sisn.com	SMTP/POP3
TC-EmailServer	TC	DHCP	Email	sisn.com	SMTP/POP3
SL-EmailServer	SL	DHCP	Email	sisn.com	SMTP/POP3
SD-EmailServer	SD	DHCP	Email	sisn.com	SMTP/POP3
RB-EmailServer	RB	DHCP	Email	sisn.com	SMTP/POP3
AU-EmailServer	AU	9.9.4.133	Email	sisn.com	SMTP/POP3

SL-DHCPServer	SL	9.9.3.2	DHCP	—	DHCP
SD-DHCPServer	SD	9.9.3.130	DHCP	—	DHCP

6. Routing Configuration Summary

Router	Routing Type	Command	Destination	Purpose
HO-Router	Next-hop Static	ip route 9.9.2.0 255.255.255.0 9.9.4.194	TC Network	HO → TC
HO-Router	RIPv2	network 9.9.0.0 / 9.9.4.0	All RIP networks	Dynamic routing
HO-Router	Redistribute Static	redistribute static	Into RIPv2	Inject static into RIP
TC-Router	Default Static	ip route 0.0.0.0 0.0.0.0 9.9.4.193	All networks	TC → HO (default)
SL-Router	Recursive Static	ip route 9.9.2.0 255.255.255.0 9.9.4.197	TC Network	SL → TC via HO
SL-Router	RIPv2	network 9.9.3.0 / 9.9.4.0	All RIP networks	Dynamic routing
SD-Router	Recursive Static	ip route 9.9.2.0 255.255.255.0 9.9.4.205	TC Network	SD → TC via SL
SD-Router	RIPv2	network 9.9.3.128 / 9.9.4.0	All RIP networks	Dynamic routing
RB-Router	Exit-Interface Static	ip route 0.0.0.0 0.0.0.0 Serial0/3/0	All networks	RB → HO (exit-if)
RB-Router	RIPv2	network 9.9.4.0	All RIP networks	Dynamic routing
AU-Router	Floating Static	ip route 0.0.0.0 0.0.0.0 Serial0/3/1 121	All networks	Backup via RB (AD=121)
AU-Router	RIPv2	network 9.9.4.128 / 9.9.4.0	All RIP networks	Primary via SD

DHCP Server Configuration (SL and SD)

Unlike TC and RB which use their routers as DHCP servers, the Software Lab (SL) and Support Division (SD) units utilize dedicated DHCP servers for dynamic IP address assignment. The SL-DHCPServer (statically assigned IP: 9.9.3.2) is configured with pool name SL-POOL, serving addresses starting from 9.9.3.10 with a subnet mask of 255.255.255.128, default gateway 9.9.3.1, and DNS server 9.9.0.6. Similarly, the SD-DHCPServer (statically assigned IP: 9.9.3.130) is configured with pool name SD-POOL, serving addresses starting from 9.9.3.140 with a subnet mask of 255.255.255.128, default gateway 9.9.3.129, and DNS server 9.9.0.6. Both servers have their DHCP service enabled through the Cisco Packet Tracer server GUI. Successful DHCP operation was verified by confirming that client PCs in both the SL and SD networks received their IP addresses

automatically, as reflected in their IP configuration panels showing DHCP-assigned addresses within the expected ranges.

DHCP

Interface FastEthernet0 Service On Off

Pool Name SL-POOL

Default Gateway 9.9.3.1

DNS Server 9.9.0.6

Start IP Address : 9 9 3 10

Subnet Mask: 255 255 255 128

Maximum Number of Users : 118

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
SL-POOL	9.9.3.1	9.9.0.6	9.9.3.10	255.255...	118	0.0.0.0	0.0.0.0

DHCP

Interface FastEthernet0 Service On Off

Pool Name SD-POOL

Default Gateway 9.9.3.129

DNS Server 9.9.0.6

Start IP Address : 9 9 3 140

Subnet Mask: 255 255 255 128

Maximum Number of Users : 80

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	9.9.3.129	9.9.0.6	9.9.3.132	255.255…	124	0.0.0.0	0.0.0.0
SD-POOL	9.9.3.129	9.9.0.6	9.9.3.140	255.255…	80	0.0.0.0	0.0.0.0

DNS and Web Server Verification

DNS was configured on HO-DNS (9.9.0.6) to resolve www.sisn.local to HO-WebServer (9.9.0.7).

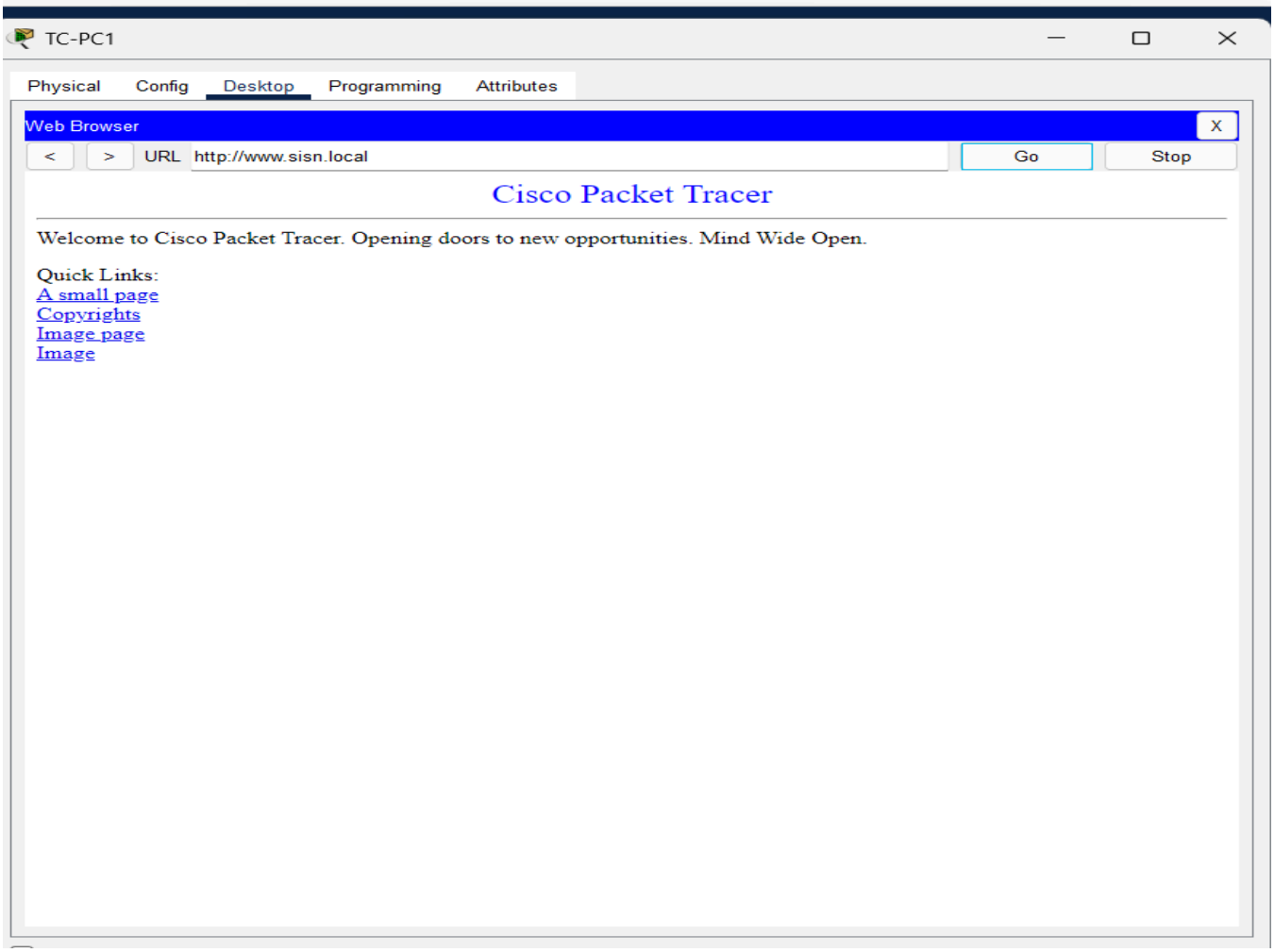
Test performed: Opened web browser on TC-PC1 and navigated to http://www.sisn.local

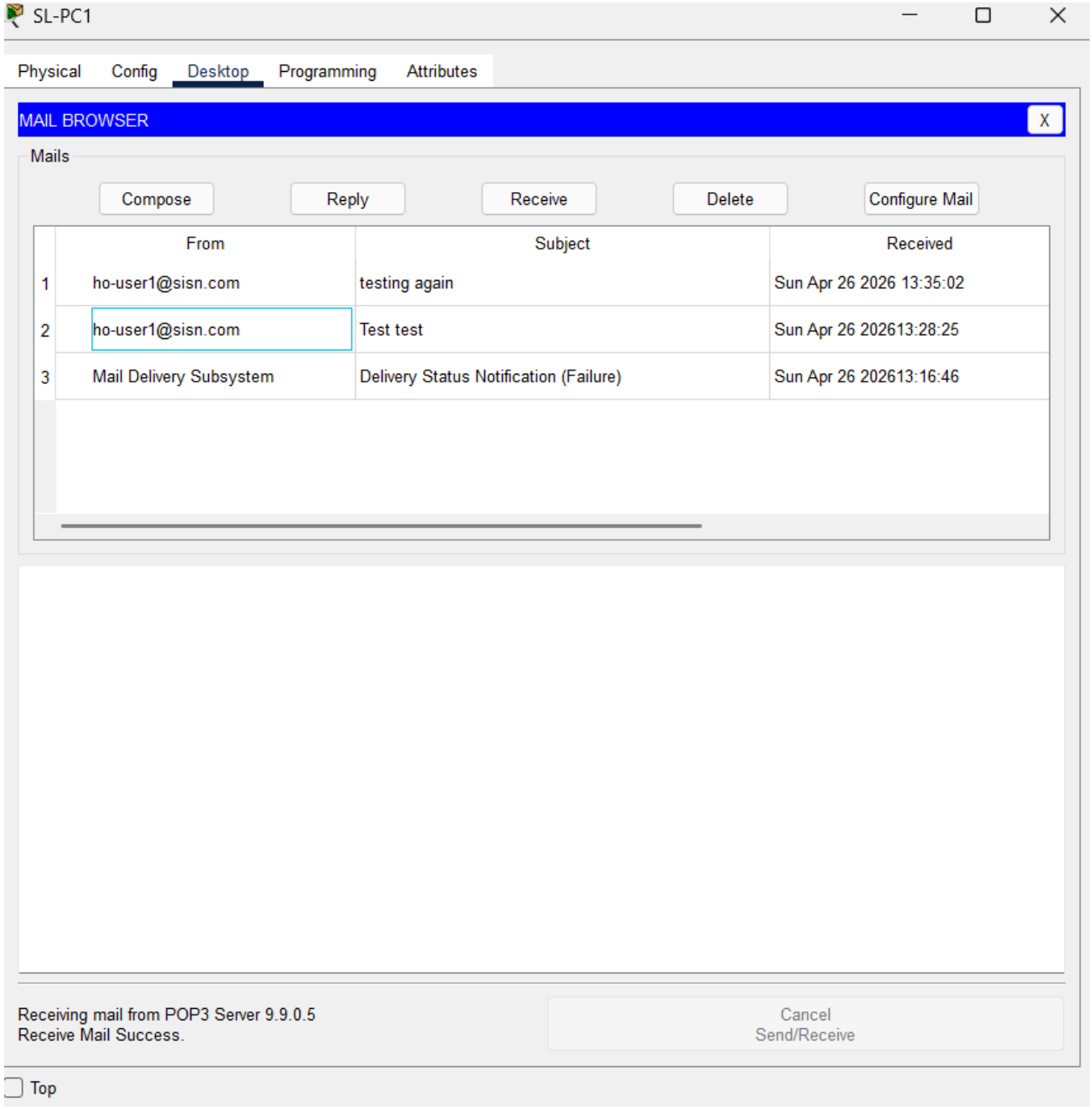
Result: Website loaded successfully confirming DNS resolution and web server functionality across the network.

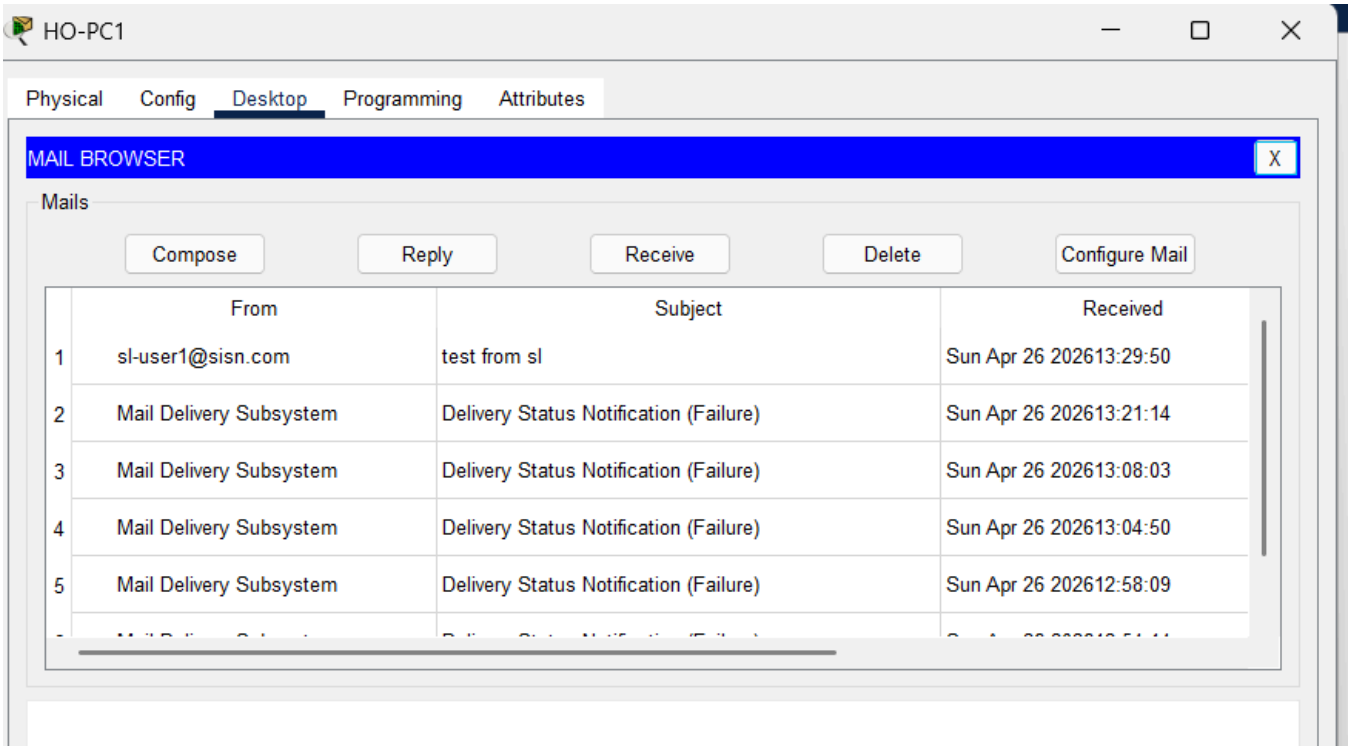
Email Communication Verification

Email servers were configured on all six units using SMTP and POP3 protocols. Email communication between HO and SL was specifically tested and verified.

Test	From	To	Result
Test 1	ho-user1@sisn.com	sl-user1@sisn.com	Received successfully
Test 2	sl-user1@sisn.com	ho-user1@sisn.com	Received successfully


TC-PC1
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://www.sisn.local
Go
Stop
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image


SL-PC1
Physical
Config
Desktop
Programming
Attributes
MAIL BROWSER
X
Mails
Compose
Reply
Receive
Delete
Configure Mail
From
Subject
Received
1
ho-user1@sisn.com
testing again
Sun Apr 26 2026 13:35:02
2
ho-user1@sisn.com
Test test
Sun Apr 26 202613:28:25
3
Mail Delivery Subsystem
Delivery Status Notification (Failure)
Sun Apr 26 202613:16:46
Receiving mail from POP3 Server 9.9.0.5
Receive Mail Success.
Cancel
Send/Receive
Top



Fault Tolerance Test 1 — HO to SL Link Failure

Objective: Verify that traffic reroutes through RB when the direct HO-SL link fails.

Procedure:

1. Recorded normal route table on HO-Router showing direct path to SL via Serial0/3/1
2. Shut down HO-Router Serial0/3/1 interface (HO-SL link)
3. Waited for RIPv2 to reconverge (approximately 30 seconds)
4. Pinged SL-Router (9.9.3.1) from HO-Router

Result: PASSED

After reconvergence, ping to SL succeeded with 100% success rate. The route table showed the SL network (9.9.3.0/25) now being reached via RB (9.9.4.202) instead of the direct serial link which confirms that RIPv2 dynamic rerouting is working correctly.

Restoration: Interface was brought back up using no shutdown command and full connectivity was restored.

```
Router>ping 9.9.3.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 9.9.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 36/64/115 ms

Router>ping 9.9.3.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 9.9.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/44/61 ms
```

Fault Tolerance Test 2 — AU Floating Static Route

Objective: Verify that AU's floating static route via RB activates as backup when the primary path via SD fails.

Procedure:

1. Recorded normal route table on AU-Router showing primary path via SD (Serial0/3/0)
2. Verified floating static route configured with AD=121: ip route 0.0.0.0 0.0.0.0 Serial0/3/1 121
3. Shut down AU Serial0/3/0 (SD link)
4. Verified all traffic rerouted via RB (Serial0/3/1) through RIPv2
5. Confirmed floating static route configuration using show running-config

Result: PASSED

The floating static route (AD=121) is correctly configured as a backup to RIPv2 (AD=120). When the SD link was shut down, RIPv2 automatically rerouted traffic via RB. The floating static route activates only when all RIPv2 paths fail, as designed.

Packet Tracer removes static routes from the routing table when the exit interface is administratively shut down, this is a known simulator limitation. The floating static route is correctly configured and verified via show running-config output.

Restoration: Interface was restored using no shutdown command.

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed sten
Router#show running-config | include ip route
ip route 0.0.0.0 0.0.0.0 Serial0/3/1 121
Router#
```

Fault Tolerance Test 3 — SD to TC Recursive Static Route

Objective: Verify that SD can still reach TC when the direct SD-SL serial cable is removed, using the recursive static route.

Procedure:

1. Verified normal connectivity from SD-Router to TC (9.9.2.1) — ping success 5/5
2. Verified recursive static route: ip route 9.9.2.0 255.255.255.0 9.9.4.205
3. Removed direct serial cable between SD-Router and SL-Router
4. Pinged TC-Router (9.9.2.1) from SD-Router again

Result: PASSED

After removing the direct SD-SL cable, SD-Router maintained connectivity to TC network via the Core Switch path (GigabitEthernet0/1 via 9.9.4.209). The recursive static route is correctly configured and connectivity to TC was maintained through the alternate path, demonstrating the network's redundancy.

Restoration: SD-SL serial cable was reconnected and clock rate was reconfigured on the DCE side.

```
Router#ping 9.9.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 9.9.2.1, timeout is 2 seconds:
!.!.!
Success rate is 60 percent (3/5), round-trip min/avg/max = 2/38/78 ms

Router#show ip route 9.9.2.0
Routing entry for 9.9.2.0/24
Known via "rip", distance 120, metric 2
  Redistributing via rip
  Last update from 9.9.4.209 on GigabitEthernet0/1, 00:00:25 ago
  Routing Descriptor Blocks:
  * 9.9.4.209, from 9.9.4.209, 00:00:25 ago, via GigabitEthernet0/1
      Route metric is 2, traffic share count is 1

Router#show running-config | include ip route
ip route 9.9.2.0 255.255.255.0 9.9.4.205
Router#
```

Assumptions

1. **Device Count** : The actual number of end devices connected in the simulation is kept minimal (2 PCs, 1 printer, 1 email server per unit) for practical demonstration purposes, though the subnets are sized to accommodate the full device counts specified (HO: 300, TC: 150, SL: 120, SD: 100, RB: 80, AU: 50).
2. **Serial Link Clocking** : All serial links use a clock rate of 64000 bps. The DCE side of each serial connection is assumed to be on the router that has the `clock rate` command configured.
3. **DNS Resolution** : All devices across the network are assumed to use the central DNS server at 9.9.0.6 (HO-DNS) for domain name resolution, including devices in remote units that receive it via DHCP.
4. **Email Protocol** : SMTP is assumed for sending and POP3 for receiving emails across all email servers. All email accounts use the domain sisn.com.
5. **RIPv2 Convergence Time** : A convergence delay of approximately 30 seconds is assumed acceptable for RIPv2 to recalculate routes during link failures, which is standard behavior for RIP.
6. **Static IPs for Servers** : All servers (DNS, Web, Email, DHCP) are assumed to require static IP addresses to ensure consistent accessibility across the network.
7. **Packet Tracer Limitations** : Cisco Packet Tracer is assumed as a sufficient simulation environment, acknowledging that certain real-world behaviors (such as floating static route visibility when an interface is shut down) may differ from physical hardware.